A forced prime condition and an explicit quartic in 392 variables

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Abstract

We specify an integer quartic in 392 total variables by a complete coefficient file and a quadratic system of 364 equations. Its rational input projection is the integers under the arithmetic input contract stated below. The new argument replaces a three-variable size condition by a single positive rational parameter with a uniformly large odd valuation at a fixed prime. The scalar local test then provides both the required lower bound on an integer index and an independent exclusion of the exceptional auxiliary elliptic points. Soundness and completeness are proved separately. The polynomial has 6204 nonzero ordinary monomials, content one, and coefficient height less than $2^{1791357}$. Exact construction data and a separate verification program accompany the proof.

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1 Statement and scope

For $P \in \mathbb{Z}[X, Y_1, \dots, Y_m]$, its rational input projection means

$$\{x \in \mathbb{Q} : \text{there exist } y_1, \dots, y_m \in \mathbb{Q} \text{ with } P(x, y_1, \dots, y_m) = 0\}.$$

A total variable count includes the input X. Coefficient height means the largest absolute value of a coefficient after all fixed integers have been substituted and equal coordinate monomials have been collected.

Theorem 1.1. *The files* `quadratic-system392.json` *and* `polynomial392.json` *specify 364 integer quadratic polynomials Q_j in 392 variables, and the polynomial $P = \sum_j Q_j^2$. The degree of P is four, its content is one, and it has 6204 nonzero coordinate monomials. Its largest coefficient has 1791357 binary digits. Under the arithmetic input contract in Section 2, its rational input projection is $\mathbb{Z}$.*

The coefficient assertions are exact finite algebraic statements. The projection assertion is a conditional theorem: the computational programs do not establish the arithmetic hypotheses. The changes made here do not assert a bijection with all witness assignments of the starting system. They prove a sound integer extraction and construct witnesses for every integer input.

2 Arithmetic inputs and the retained system

Put

$$K = \mathbb{Q}(\rho), \quad \rho^2 = 2, \quad d_0 = 3 - \rho, \quad D = d_0^2, \qquad E : y^2 = x^3 - Dx.$$

The fixed points are

$$R_0 = (1 + \rho, 2\rho), \qquad P = 2R_0 = \left(\frac{57 - 28\rho}{8}, \frac{-660 + 417\rho}{32}\right).$$

For a point in its domain, write $f = 1/x$ and $z = -x/y$, extended by $f(O) = z(O) = 0$ near the identity. Heights are absolute logarithmic Weil heights. At an unramified place above p, the valuation is normalized by $v(p) = 1$.

The arithmetic input contract consists of the following assertions, with the original sources and applications identified in [1, Sections `subsec:v12-inputs` and `sec:v13`].

1. The equality $2E(K) = \mathbb{Z}P$ holds, and $h(x([n]P)) \geq n^2 - 6$ for every nonzero integer n.

2. For $t = u/v$ in lowest terms, $v > 0$, outside $\{0, 1, 2401, -343/32\}$, let $M_5(t)$ be the product of primes $p > 76176$ for which a positive odd valuation occurs in at least one of

   $$v, \quad u, \quad u - v, \quad u - 2401v, \quad 32u + 343v.$$

   The contact-height inequality is

   $$h(t) \leq \mathcal{H} M_5(t)^{19}, \qquad \mathcal{H} = 2^{110} 210^{1800} \left(\prod_{p \leq 76176} p\right)^{19} + 128.$$

   The exact numerical bound used here is $\mathcal{H} < 2^{2093448}$.

3. The pole equations in the source, with input b and $B = 17b$,

   $$U = b\frac{1 - bt^2}{1 + bt^2}, \quad h_0 = e(B - U^2), \quad p^2 = B(1 - e)(1 + 3e), \quad Bq^2 = (s^2 - Bh_0)(s^2 + 3Bh_0),$$

   exclude negative odd $v_\ell(b)$ when $\ell \notin \{2, 3, 17\}$. They admit the specialization $t = 0$ if $0 < b < 17$, the reduced denominator of b is a square, b is a square in each of $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, and its valuations there are at least $3, 1, 2$, respectively. We use the exact normalized seven-coordinate form of these equations in the specified starting system.

4. The source simultaneous prime-pattern and conjugate reduction-order conditions can be realized with arbitrary fixed odd avoidance modulus. In the notation of that source, selected primes are 1 modulo $8 \cdot 3 \cdot 17$, the orders satisfy $v_2(d_-) = 6$, $v_2(d_+) = 2$ and $\gcd(d_-(p), d_+(p')) = 4$. The required fixed congruences and real approximation conditions are simultaneous.

As in the quartic source, we retain the following quantitative form of the last input for the separation step. For the fixed distinct coefficients c_j, the admissible patterns $p_j = u + c_j w^2$ with $N/2 \leq u \leq N$ and $1 \leq w \leq \tau(N)\sqrt{N}$ have total weight at least $c_0 N^{3/2}\tau(N)$, individual weights are $O((\log N)^k)$, and $\tau(N)$ may decrease arbitrarily slowly after all fixed congruences and avoidance conditions are imposed. This is an arithmetic premise; finite coefficient checks do not prove it. The precise source locations are Proposition `prop:v5-artin-patterns`, the final diagonal argument in `subsec:v5-diagonal`, and the separation application in the retained Version 9 portion of [1].

2.1 The unchanged comparison interface

The starting quadratic system is the 394-variable system of [1], at the commit specified in Section 7. We retain all its point charts, comparison equations, three height tests, and their normalized poles. The following are the properties of that unchanged part used here. Their proofs and precise local applications are given in the Version 13 sections on reciprocal charts, the enlarged H domain, scaled charts, and the two final height contradictions.

The four root points are

$$Q_1 = [n]P, \quad S = Q_2 = [a]P, \quad Q_3 = [t]P, \quad Q_4 = [b]P, \qquad n, a, t, b \text{ nonzero odd integers.}$$

The guarded reciprocal subtraction chart gives

$$G = S - P = [s]P, \qquad s = a - 1 \neq 0.$$

The auxiliary finite point is $H = [m]R_0 + T$ with $m \in \mathbb{Z}$ and $T \in E[2]$. Its retained chart is

$$x(H)f(H) = 1, \qquad H = (x, xS_H), \qquad S_H^2 = x - Df(H).$$

It follows immediately, before assigning meaning to any ratio slot, that $f(H) \neq 0$ and $q = \mathrm{Norm}_{K/\mathbb{Q}}(f(H)) \neq 0$.

After excluding

$$E[2] \ \cup \ (P + E[2]) \ \cup \ (-P + E[2]), \tag{1}$$

the retained ratio equations identify their scalar slots with

$$\frac{z(G)}{z(H)} = z_0 + z_1\rho.$$

The proof also identifies the primary and alternate comparison slots, including the identity branches of their numerators. It uses the integral group identities

$$[m]U_i = [2u_i]H, \qquad [m-2]V_i = [2(u_i - n_i)](H - P),$$

so that $h = m/2$ is only a rational index; no fractional group multiple is asserted. The resulting order-forty tests make the three contact radicals divide

$$F = t^2 - a(a-2)tb + (a-2)^2b^2 - 1 \tag{2}$$

to order forty. The order-ten tests make them divide the nonzero reduced numerator of $n - \beta$ to order ten, if that difference is nonzero. The common height consequence is

$$B_*^2 - 6 < 7\mathcal{H}|N|^{19/k}, \qquad B_* = \max(|n|, |a|, |t|, |b|), \tag{3}$$

whenever their kth powers divide $N \neq 0$. The three scales remain $1, C_0^2, (1 + C_1^2)^2$ for $\prod_i x(Q_i) = C_0 + C_1\rho$. None of these equations, scales, or comparison orders is changed below.

For completeness the source construction permits any fixed finite list of formal-depth conditions on H and G, imposed before the later contact primes are chosen. Let σ be conjugation on K, and put

$$G_{\mathrm{real}} = \overline{\{([n]P, ([n]P)^\sigma) : n \in \mathbb{Z}\}} \subset E(\mathbb{R}) \times E^\sigma(\mathbb{R}).$$

Write G_{real}^0 for its identity component and $m_0 = |G_{\mathrm{real}}/G_{\mathrm{real}}^0|$. A progression with residue and modulus divisible by m_0 has dense positive tail in G_{real}^0. Thus the prescribed finite conditions and small real values of $f(H)$ and $z(G)/z(H)$ can hold at the same global points. The fixed modulus can include any further finite odd support. This simultaneous version, rather than separate choices at different places, is used in Section 6.

3 A positive parameter with a forced odd valuation

Define

$$\beta(X) = 51(2X+1)^2 + 2504, \tag{4}$$
$$r(X) = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1). \tag{5}$$

For every rational X, all five factors are positive. In particular $r(X) > 0$ without any integrality assumption.

Lemma 3.1. *The integer $\ell = 80989$ is prime, is greater than 76176, and is 1 modulo 17. For every $X \in \mathbb{Q}$,*

$$v_\ell(r(X)) = \begin{cases} 0, & v_\ell(X) \geq 0, \\ 10v_\ell(X), & v_\ell(X) < 0. \end{cases}$$

Proof. Trial division through $\lfloor\sqrt{80989}\rfloor = 284$ certifies primality. For each of the five contact roots θ, Euler's criterion gives

$$\left(\frac{(\theta - 2504)/51}{\ell}\right) = -1.$$

The exact residues and modular powers are listed in the verification file and independently recomputed by the checker. Hence, if X is ℓ-integral, $\beta(X)$ misses all five roots of the product modulo ℓ, and r is a unit. If $v_\ell(X) < 0$, the leading term of β gives $v_\ell(\beta) = 2v_\ell(X)$. Every contact factor then has that valuation, proving the second assertion. □

Lemma 3.2. *Let $N \geq 1$ be an integer and put*

$$A = \ell^{2N+1}, \qquad R = \frac{Ar}{1+\ell^2 r}, \qquad c = R + 4896.$$

For every rational X, $R, c > 0$ and

$$v_\ell(R) \in \{2N+1, 2N-1\}. \tag{6}$$

Every prime occurring in $M_5(\beta)$ is different from ℓ and has positive odd valuation on R; at such a prime c is a unit. For an integer X,

$$v_2(R) = 2, \quad v_3(R) = 1, \quad v_{17}(R) = 0, \qquad c/R \in \mathbb{Q}_p^{\times 2} \quad (p = 2, 3, 17). \tag{7}$$

Proof. The denominator $1 + \ell^2 r$ is positive. If $v_\ell(r) = 0$, it is a unit. Otherwise $v_\ell(r) \leq -10$, so its valuation is $2 + v_\ell(r)$. These two cases give (6).

The reduced denominator of β has even valuation at every prime larger than 76176. Therefore its contact radical only consumes positive odd valuations at the four finite roots defining M_5. Distinct roots of the five factors in (5) cannot coincide modulo a prime larger than 76176: the relevant nonzero differences have no such prime divisor. At a consumed prime, consequently, $v_p(r) > 0$ is odd. Lemma 3.1 excludes $p = \ell$. Both A and $1 + \ell^2 r$ are then units, so $v_p(R) = v_p(r)$ and c is a unit, since every prime factor of 4896 is 2, 3, or 17.

For integral X, the odd integer $\tau = 2X + 1$ gives $v_2(r) = 2$, $v_3(r) = 1$ and $r \equiv 15 \pmod{17}$. These follow directly by reducing the five displayed factors modulo $8, 9, 17$, respectively. The new denominator is a unit at $2, 3$, and is 16 modulo 17. The numerator multiplier is a unit at all three primes. Thus the valuations of R are as stated. Finally

$$v_2(4896/R) = 3, \qquad v_3(4896/R) = 1, \qquad v_{17}(4896/R) = 1.$$

The unit $1 + 4896/R$ is therefore a local square: it is 1 modulo 8 at two and 1 modulo the maximal ideal at the two odd primes. □

4 Local extraction before interpreting the ratio slots

The retained scalar norm equations, with their parameter changed from r to R, are

$$B_s = \frac{z_0^2 + Rz_1^2}{R^{32}}, \qquad W = \frac{B_s^2 + Rq^2}{R^{32}}, \qquad \mathfrak{b} = (R + 4896)RW^2. \tag{8}$$

Initially z_0, z_1 in this display mean the stored rational slots. Their ratio interpretation is not used in the next lemma. Since $q \neq 0$ and $R > 0$, we have $W > 0$ and $\mathfrak{b} > 0$. Hence the normalized pole has a nonzero input before any logarithm denominator is interpreted.

Lemma 4.1. *Suppose $p > 76176$, $\nu = v_p(R) > 0$ is odd, and the pole equations for (8) hold. Then*

$$v_p(B_s) > \frac{63\nu}{4}, \quad v_p(q) > \frac{61\nu}{4}, \quad v_p(z_0) > \frac{191\nu}{8}, \quad v_p(z_1) > \frac{187\nu}{8}.$$

Proof. The correction $R + 4896$ is a unit. The valuation of $\mathfrak{b}$ is $\nu + 2v_p(W)$, an odd integer. The pole obstruction excludes negative values, so $v_p(W) > -\nu/2$. The summands in each of the two norm numerators have valuations of opposite parity. They cannot cancel; a zero summand is assigned valuation $+\infty$. The second equation in (8) therefore gives

$$2v_p(B_s) > \frac{63\nu}{2}, \qquad \nu + 2v_p(q) > \frac{63\nu}{2}.$$

Apply the same argument to the first numerator to obtain $2v_p(z_0) > 191\nu/4$ and $\nu + 2v_p(z_1) > 191\nu/4$. These are the asserted bounds. □

Lemma 4.2. *The equations with the forced parameter exclude every exceptional point in (1), before the ratio slots need to be interpreted.*

Proof. The points O and $(0, 0)$ are already excluded by $xf(H) = 1$. For every remaining exceptional point, its abscissa is among

$$\pm d_0, \quad c_P, \quad -D/c_P, \quad d_0\frac{c_P + d_0}{c_P - d_0}, \quad -d_0\frac{c_P - d_0}{c_P + d_0}, \qquad c_P = \frac{57 - 28\rho}{8}.$$

These are the usual translations by the three two-torsion points. The factors and their inverses are integral outside $\{2, 3, 7, 17, 41\}$, because

$$\mathrm{Norm}(d_0) = 7, \quad \mathrm{Norm}(c_P) = \frac{41^2}{64}, \quad \mathrm{Norm}(c_P - d_0) = \frac{17^2}{64}, \quad \mathrm{Norm}(c_P + d_0) = \frac{63^2}{64}.$$

Thus $f(H)$ is a unit at every place above ℓ, and q is an ℓ-unit. This contradicts Lemma 4.1 and (6). No meaning of z_0, z_1 was used. □

Proposition 4.3. *Every solution has a nonzero even integer $s = a - 1$ satisfying*

$$v_\ell(s) \geq L_N := \left\lfloor \frac{187(2N - 1)}{8} \right\rfloor + 1.$$

Moreover $M_5(\beta)^{24}$ divides s.

Proof. The retained reciprocal guard gives $G = [s]P$ with $s \neq 0$ even. Lemma 4.2 now permits the ratio interpretation $z_0 + z_1\rho = z(G)/z(H)$. At ℓ, positive valuation of q supplies a place at which $v(f(H)) > 0$, so H is formal there. The rational coefficient depths from Lemma 4.1 give

$$v(z(G)) \geq v(z(H)) + L_N.$$

If $e = v(x(G))$, the identity $z(G)^2 = x(G)/(x(G)^2 - D)$ shows that $e \neq 0$. The duplication formula

$$x(2G) = \frac{(x(G)^2 + D)^2}{4x(G)(x(G)^2 - D)}$$

gives $v(x(2G)) = -2v(z(G))$ for both signs of e. Hence $2G$ is formal and $v(z(2G)) = v(z(G))$. Apply the formal logarithm, an isometry at this positive depth, to the integral group identity

$$[m](2G) = [4s]H.$$

The excluded set implies $m \neq 0$. Since $m/2$ is ℓ-integral, we obtain $v_\ell(s/(m/2)) \geq L_N$ and consequently $v_\ell(s) \geq L_N$.

For a prime in $M_5(\beta)$ the same proof applies, by Lemma 3.2. Here $\nu \geq 1$ is odd, so both coefficient depths are at least 24. The formal comparison gives $v_p(s) \geq 24$ at each such prime, which proves the divisibility assertion. □

5 The numerical choice and integer extraction

We now fix

$$N = 27464, \qquad A = 80989^{54929}.$$

Then $2N - 1 = 54927$ and $L_N = 1283919$. The entirely integral certificates

$$80989^{4096} > 2^{66787}, \qquad 66787 \cdot 1283919 - 4096 \cdot 20934550 = 1181453 > 0$$

show that every nonzero index in Proposition 4.3 satisfies

$$|s| \geq 80989^{1283919} > 2^{20934550} > (80\mathcal{H})^{10}. \tag{9}$$

Since $|a| = |s + 1| \geq |s| - 1$, this implies

$$|a| > (40\mathcal{H})^{10}, \qquad \log(48\mathcal{H}) < |a|/40. \tag{10}$$

The scalar contact-height inequality and the divisibility just proved give

$$h(\beta) \leq \mathcal{H}|s|^{19/24} < |s|/80 \leq |a|/40. \tag{11}$$

The middle inequality follows from $|s|^{5/24} > (80\mathcal{H})^{50/24} > 80\mathcal{H}$.

For clarity, the elementary growth fact needed for the integer extraction is proved here, rather than included among the arithmetic inputs.

Lemma 5.1. *If integers a, t, b satisfy $|a| > 7$, $b \neq 0$, and $t^2 - a(a-2)tb + (a-2)^2b^2 = 1$, then* $\log|t| > |a| - 3$.

Proof. Set $A_0 = |a|$ and $c = \operatorname{sgn}(a)(a-2)b$. Then $t^2 - A_0tc + c^2 = 1$. The nonzero integers t, c have the same sign, and $T = |t|$, $C = |c|$ are positive solutions. Define $U_0 = 0$, $U_1 = 1$, $U_{j+1} = A_0U_j - U_{j-1}$. All positive solutions are unordered adjacent terms of this sequence: for $M \geq m > 0$, the replacement

$$(M, m) \longmapsto (m, A_0m - M), \qquad A_0m - M = (m^2 - 1)/M \in [0, m),$$

preserves the equation and descends to $(1, 0)$. Thus $C = U_j$ and $T \geq U_{j-1}$ for some $j \geq 1$. If $d = |a - 2|$, reduction of the recurrence modulo d gives $U_j \equiv j$ or $(-1)^{j-1}j$, according to the sign of a. Since $d \mid C$, we have $j \geq d \geq A_0 - 2$. The recurrence gives $U_r > (A_0 - 1)^{r-1}$ for $r \geq 2$. Hence

$$\log T > (j-2)\log(A_0 - 1) \geq (A_0 - 4)\log 7 > A_0 - 3 \qquad (A_0 \geq 8).$$

The final inequality holds at 8 and its difference increases. □

Proposition 5.2 (Soundness)**.** *Every rational solution of the new quadratic system has integer input.*

Proof. All the unchanged comparison slots have their required meaning by Lemma 4.2. If the integer F in (2) were nonzero, its order-forty divisibility and (3) would give

$$B_*^2 - 6 < 7\mathcal{H}(5B_*^4)^{19/40}.$$

As $B_* > 4$, this implies $B_*^{1/10} < 40\mathcal{H}$, contrary to $B_* \geq |a|$ and (10). Thus $F = 0$, and Lemma 5.1 yields $\log B_* > |a| - 3$.

If $n - \beta \neq 0$, let N_0 be its nonzero reduced numerator. The source comparison gives order-ten divisibility and $|N_0| \leq 2e^{h(\beta)}B_*$. Its height consequence is

$$\log B_* < 10\log(48\mathcal{H}) + 19h(\beta) < 29|a|/40 < |a| - 3,$$

where (10), (11), and the enormous lower bound on $|a|$ justify the last two inequalities. This is impossible. Hence $n = \beta$.

Finally, write $2X + 1 = u/v$ in lowest terms with $v > 0$. Integrality of $51u^2/v^2 + 2504$ implies $v^2 \mid 51$, so $v = 1$ because 51 is squarefree. Since $n = \beta$ is odd, u is odd. Therefore $X = (u - 1)/2 \in \mathbb{Z}$. □

6 Simultaneous completeness

Fix an integer input X. The numbers R, c are now fixed positive rationals satisfying (7). Let S contain $2, 3, 17$ and every prime with nonzero valuation on cR or R. This is a finite set, fixed before the later prime patterns are chosen.

Lemma 6.1. *The same global auxiliary points can be chosen to make*

$$0 < \mathfrak{b} < 17,$$

while $\mathfrak{b}$ *has square reduced denominator, is a local square at* $2, 3, 17$*, and has valuations there at least* $3, 1, 2$*. These requirements are compatible with every retained comparison and prime-selection condition.*

Proof. At a fixed finite place all coefficients in (8) are fixed. If the two rational coefficients of $f(H)$ and those of $z(G)/z(H)$ tend to zero, then q, B_s, W tend to zero. In valuation terms, if both pairs have depth at least t and $\alpha = v_p(R)$, then

$$v_p(q) \geq 2t, \qquad v_p(B_s) \geq 2t + \min(0, \alpha) - 32\alpha,$$

and

$$v_p(W) \geq \min\{2v_p(B_s), \alpha + 2v_p(q)\} - 32\alpha.$$

These lower bounds tend to infinity with t. Choose sufficiently large depths at every prime in S that $\mathfrak{b}$ is integral there and has the three required special lower bounds. Away from S its valuation is $2v_p(W)$, which is even. Thus the reduced denominator of $\mathfrak{b}$ is a square. Its local squareclasses at $2, 3, 17$ equal those of cR, which are squares by (7).

These local conditions must be achieved together with the real condition. In the paired real orbit closure, first choose a positive index h_0 divisible by $64m_0$, in a sufficiently divisible progression, so that $H_0 = [h_0]P$ has all selected formal depths and is close to both real identities. Then $q(H_0)$ is arbitrarily small. Fix this point. Next choose a positive index s divisible by 6 in a sufficiently divisible progression, arbitrarily large and with $G = [s]P$ close to both real identities. As $z(H_0)$ is nonzero, the two embeddings of $z(G)/z(H_0)$, and therefore its two rational coefficients, are arbitrarily small. The displayed norm tree is continuous and tends to zero at

these values. Choose the points so that $0 < \mathfrak{b}(H_0, G) < 17/4$ and all finite conditions hold. Strict positivity follows from $q(H_0) \neq 0$.

Keep G fixed. Real continuity supplies a neighborhood of H_0 with $0 < \mathfrak{b} < 17$, and finite continuity supplies a fixed congruence $h \equiv h_0 \pmod{L}$ preserving all selected depths. Choose $64m_0 \mid L$. The remaining root labels and comparison scales are fixed before the new contact primes. A smaller nonempty real open set with compact closure avoids all comparison poles; their bounds on that closure are consequently fixed as well.

The retained prime-selection and separation argument allows this additional fixed modulus. Put

$$\Pi = \prod_p p, \qquad M_\pm = \operatorname{lcm}_p(d_\pm(p)/4),$$

where the products and least common multiples run over the selected primes. The order and avoidance conditions give $\gcd(M_-, M_+) = 1$, $\gcd(M_-, L) = 16$, and $\gcd(M_+, L) = 1$. The separation step gives $\gcd(\Pi, M_+) = 1$, and the selected primes avoid L. Since $64 \mid h_0$, the following congruences are compatible:

$$h \equiv 0 \pmod{\Pi^{19} M_-}, \qquad h \equiv 1 \pmod{M_+}, \qquad h \equiv h_0 \pmod{L}.$$

The resulting progression has residue and modulus divisible by m_0, so its positive tail meets the selected real open set. A single such final H satisfies the finite depths, the real bound, all comparison domains, and the contact requirements. This proves simultaneity. □

Proposition 6.2 (Completeness)**.** *Every integer input admits a rational solution of the new system.*

Proof. Use the unchanged source construction of the roots, charts, and comparison points with the finite conditions of Lemma 6.1. The pole-realization premise supplies fresh scalar pole witnesses with preparation specialized to zero. Its exact normalized chart permits zero private witnesses and divides only by the nonzero semantic pole input. Thus all five normalized pole equations hold. The guard on the reciprocal scalar point is also available. Indeed, write $s = 2 \cdot 3^k u$ with $3 \nmid u$. Direct duplication of the specified point P gives

$$z(2P)/3 \equiv 2 + \rho \pmod{3}.$$

The integral odd formal multiplication series has linear term nz and terms of degree at least three thereafter. At positive depth, iteration of multiplication by 3 and then by u therefore gives

$$z([s]P)/3^{k+1} \equiv u(2 + \rho) \pmod{3}.$$

Using $f = z^2(1 - Df^2)$ and $(2 + \rho)^2 \equiv \rho \pmod{3}$, we find that the ρ-coefficient of $f([s]P)$ has valuation exactly $2 + 2k$. If $f([s]P) = g^2$, this coefficient is $2g_0g_1$. It is nonzero, so g_0^{-1} exists. This argument also covers negative u and needs no deleted anchor condition.

Assign the new coordinate its uniquely determined value $R = Ar/(1 + \ell^2 r)$, and compute its five successive squares. The scalar equations hold by definition. The deleted three private anchor witnesses have no other occurrences. All remaining retained witnesses are supplied by the same simultaneous construction. □

Because rational squares are nonnegative, $\sum_j Q_j^2 = 0$ over $\mathbb{Q}$ is equivalent to all the equations $Q_j = 0$. Propositions 5.2 and 6.2 therefore prove the projection part of Theorem 1.1.

7 Exact quadratic implementation

The starting file is the original-byte `candidate394-ir.json` at upstream commit

`5bbd74f6a479776a332cbe036809d5dddae63720`.

Its SHA-256 digest is

`7eb02f9913b55d60a790a287616f1bb80d9f93523e9f11e12481c9ab955746a5`.

The complete starting coefficient table is separately supplied and is independently reconstructed from this system before the new computation.

We use the source coordinate numbers v_i only to make the exact changes unambiguous; the final files renumber the coordinates from 1 through 392, with the input at coordinate 1. The raw contact in the retained source gates is the affine expression

$$\begin{aligned} r = & -105275102108 - 8412883296v_0 - 8412883296v_{248} \\ & - 21050600v_{487} - 7629700v_{488} - 45771136v_{489}. \end{aligned}$$

The source equations numbered $200, 453, 454, 455$ imply that this expression equals (5). The independent checker proves the identity as an exact univariate polynomial of degree ten.

Delete the source coordinates $55, 56, 57$ and their sole equation, numbered 504. Add one coordinate $v_{702} = R$ and the quadratic equation

$$R + \ell^2 rR - Ar = 0. \tag{12}$$

It is quadratic because the displayed source expression for r is affine in retained coordinates. The denominator is always positive, so (12) has the unique intended rational solution. Replace exactly the following four source rows:

Source row	New equation
560	$v_{600} - R^2 = 0,$
566	$v_{604}v_{540} - v_{508}^2 - Rv_{539} = 0,$
512	$v_{604}v_{542} - v_{540}^2 - Rv_{541} = 0,$
492	$(v_{600} + 4896R)v_{525} + 82479248v_{526} = 0.$

The retained power rows give $v_{600} = R^2$, $v_{601} = R^4$, $v_{602} = R^8$, $v_{603} = R^{16}$ and $v_{604} = R^{32}$. The retained row 565 gives $v_{539} = v_{509}^2$, and row 511 gives $v_{541} = q^2$ through the exact identity

$$q^2 = (f_0^2 + 2f_1^2)^2 - 8(f_0 f_1)^2.$$

If J is the positive integer coefficient in row 491, $Jv_{525} - v_{542}^2 = 0$, then the actual semantic pole input is $-82479248Jv_{526} = (R + 4896)RW^2$. This sign and fixed scale are checked from the literal rows.

The guard $v_{543}v_{549} = 1$ is retained. In particular, the nonzero scalar index has not been replaced by an implicit assumption. Every other coordinate and residual is unchanged. The final count is

$$394 - 3 + 1 = 392, \qquad 364 - 1 + 1 = 364.$$

For each rational-coefficient row, clear denominators and divide by the gcd of its integer coefficients. Square these primitive rows and add. The resulting polynomial is the supplied P. Its symbolic coefficient table uses $A = 80989^{54929}$ as a fixed integer, never as an extra variable.

8 Independent computational verification

The producer `build_quartic392.py` constructs the complete system and polynomial. The separate program `verify_quartic392.py` imports no producer or earlier checker. It reconstructs the upstream 394-variable sum of squares and matches its public coefficient table. It then checks the full changed system, including every unchanged row, all coordinate incidences, the affine raw contact, the normalization and inverse guard. Finally it expands all 364 squares and compares every coefficient.

The new table has 6225 terms when powers of the fixed parameter are listed separately. After those powers are substituted and equal coordinate monomials are collected, it has 6204 nonzero monomials. These two counts are intentionally distinct. A parameter-independent coefficient equal to one certifies content one; a nonzero quartic monomial certifies the exact degree.

The maximum coefficient is the constant coefficient. If C_{const} is the sum of the squared constant terms of the 359 unchanged primitive residuals, then it is exactly

$$1108284712384982604366 4 \cdot 80989^{109858} + C_{\mathrm{const}}, \qquad 0 < C_{\mathrm{const}} < 2^{1514}.$$

The full integer C_{const} is included in `construction-report392.json`; its definition here also specifies it without abbreviating decimal digits. The independent checker compares all specialized coefficients as exact integers and proves that the maximum has 1791357 binary digits.

The same program checks trial division for 80989, all five Euler criterion calculations, the local input residues, and the integral height margin. Its receipt is `independent-verification392.json`. Run from the construction directory:

```
python build_quartic392.py
python verify_quartic392.py
```

These checks establish the finite algebra and numerical certificates. The arithmetic hypotheses and the simultaneous construction remain the mathematical inputs and arguments stated in this paper.

References

[1] E. Glazer, *An explicit quartic defining the integers in the rationals*, Version 13, repository manuscript and coefficient data, commit `5bbd74f6a479776a332cbe036809d5dddae63720`, https://github.com/elliotglazer/diophantine-z-q. The Version 13 addition and the separate mathematical composition reviews identify the enlarged auxiliary-point domain, scalar norm calculation, pole normalization, and simultaneous completeness argument used here.

[2] J. H. Silverman, *The Arithmetic of Elliptic Curves*, second edition, Graduate Texts in Mathematics 106, Springer, 2009, Chapter IV, especially Theorem 6.4.